TOPOLOGICAL INSULATORS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on Topological Insulators

Question 1

10.0 points

Analyze the band structure diagram of a topological insulator shown in the diagram below, which features band inversion between the bulk valence and conduction bands. Explain how this inversion leads to a non-trivial topology.

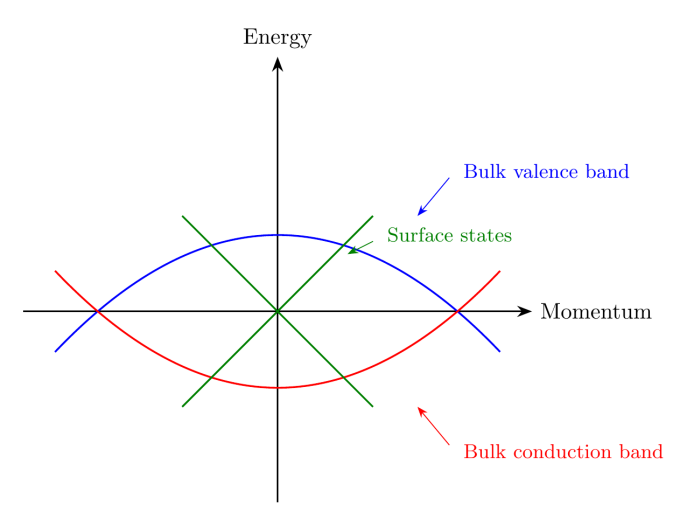

Fig. 1. A band structure diagram of a topological insulator. The diagram should show the bulk valence and conduction bands with a clear band inversion. The valence band should be depicted as parabolic and the conduction band as parabolic, with the two bands crossing each other to indicate inversion. Additionally, include linear Dirac surface states that cross through the gap created by the band inversion, highlighting the non-trivial topology. Label the valence band, conduction band, and surface states clearly. IMPORTANT: The diagram is missing required labels. The diagram is missing the. Add each missing label as visible text at the appropriate location in the diagram. IMPORTANT correction needed: The diagram contains the label "Band inversion",. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 2 10.0 points

Interpret the bulk-boundary correspondence as depicted in the surface state diagram of a topological insulator in the diagram below. Describe how the surface states are protected by topology.

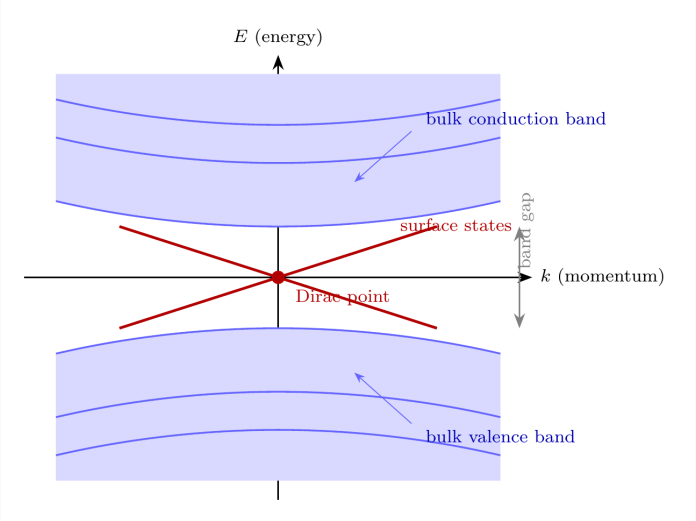

Fig. 2. A surface state diagram of a topological insulator. The diagram shows bulk conduction and valence bands as parabolic curves with a gap between them. Linear Dirac surface states are depicted as lines crossing through the bulk band gap, touching at the Dirac point. The axes are labeled with energy (E) on the vertical axis and momentum (k) on the horizontal axis. The diagram illustrates the concept of bulk-boundary correspondence, highlighting how the surface states are protected by topology. IMPORTANT: The diagram is missing required labels. The diagram has several overlapping and confusing labels. Add each missing label as visible text at the appropriate location in the diagram. IMPORTANT: The diagram is missing required labels. The diagram has a garbled and illegible label for. Add each missing label as visible text at the appropriate location in the diagram.

Question 3

10.0 points

Examine the edge state diagram shown below associated with the quantum spin Hall effect. Explain how these states allow for dissipationless edge transport.

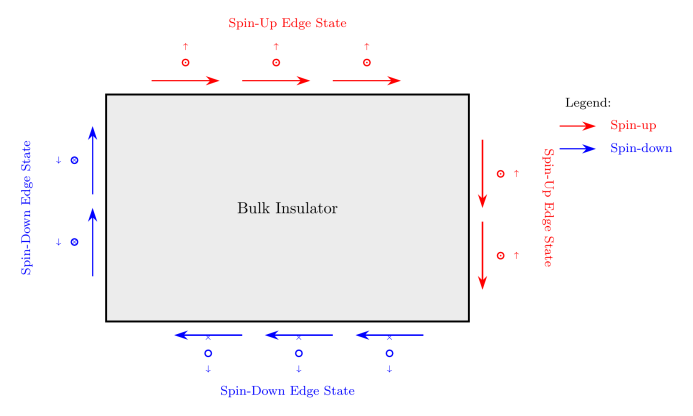

Fig. 3. A diagram showing the edge state configuration in a quantum spin Hall effect system. The diagram should include a rectangular sample with edge states depicted along the boundaries. Use arrows to indicate the direction of spin-polarized edge state transport, with one edge showing spin-up electrons moving in one direction and the opposite edge showing spin-down electrons moving in the opposite direction. Label the edges as 'Spin-Up Edge State' and 'Spin-Down Edge State'. Include a central region labeled 'Bulk Insulator' to indicate that the interior of the sample is insulating, while the edges are conductive. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram incorrectly shows spin-up and. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram incorrectly depicts the spin-polarized edge states. Regenerate using EXACTLY the values and sequences specified.

Question 4 10.0 points

Analyze the k-space structure of a 2D system with a Z2 topological invariant as depicted in the diagram below. Explain the significance of the Z2 invariant.

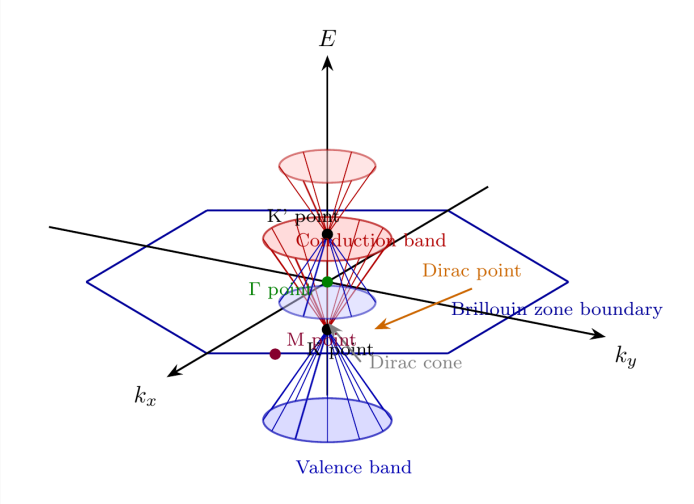

Fig. 4. CRITICAL: The previous diagram showed the wrong type. The diagram shows a gapless semimet. Regenerate from scratch showing the CORRECT diagram type. A k-space diagram of a 2D system with a Z2 topological invariant. The diagram should show the band structure with a Dirac cone at the K point, indicating the presence of a topological insulator phase. The bands should be labeled, and the Dirac cone should be clearly highlighted to show the band inversion characteristic of a Z2 topological invariant. Include axes labeled as kx and ky, and indicate the Brillouin zone boundary. IMPORTANT correction needed: The diagram contains a label "Band inversion. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 5 10.0 points

Examine the Dirac cone dispersion relation in the diagram below on the surface of a 3D topological insulator. Discuss how this structure is related to massless Dirac fermions.

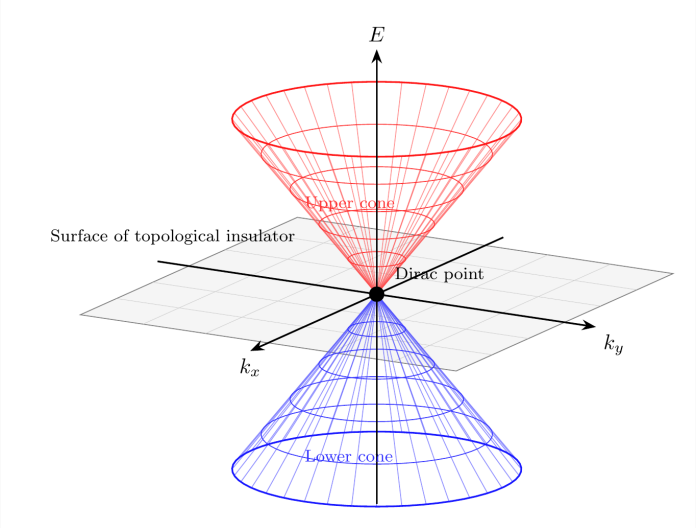

Fig. 5. A Dirac cone dispersion relation diagram on the surface of a 3D topological insulator. The diagram should show a 3D plot with a V-shaped linear dispersion relation, where the energy (E) is plotted against the momentum (k). The Dirac point, where the two cones meet, should be clearly labeled. The surface of the topological insulator should be indicated, and the linear nature of the dispersion should be emphasized to illustrate the concept of massless Dirac fermions.